

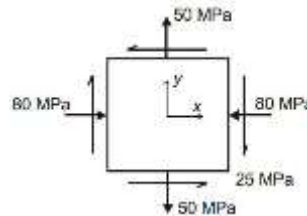
Addis Ababa University
College of Technology and Built Environment
School of Mechanical and Industrial Engineering

Machine Elements I

2025/26

Problems selected for individual assignment {2, 4, 5, 7, 8, 13 and 16}, its submission will be during next week class (March 25/2026).

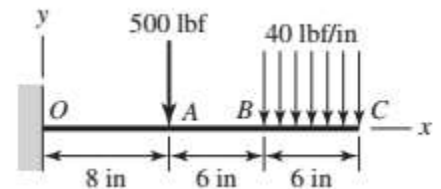
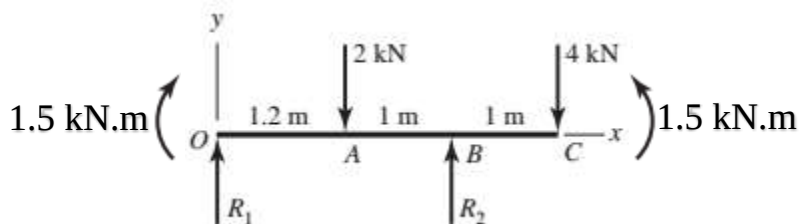
- The state of plane stress at a point is represented by the stress element below. Find the stresses on an element inclined at 35° clockwise and calculate all principal and shear stress acting on the element using both Mohr's circle and manual calculation by using simple formula and finally compare your answers? Explain also stress tensor including its representation?



- The stress state in a point of a body is given by $\begin{pmatrix} 0 & 0 & 2 \\ 3 & -4 & 0 \\ 4 & 3 & 0 \end{pmatrix} \times 10^2 \text{ MPa}$.

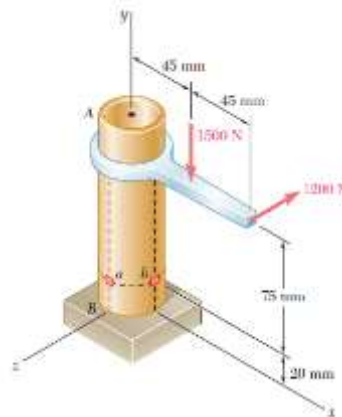
Determine the principal stress, maximum shear stress, principal axis direction angle and draw the Mohr's circle?

- For the beams shown, find the reactions at the supports and plot the shear-force and bending moment diagrams. Label the diagrams properly and provide values at all key points.

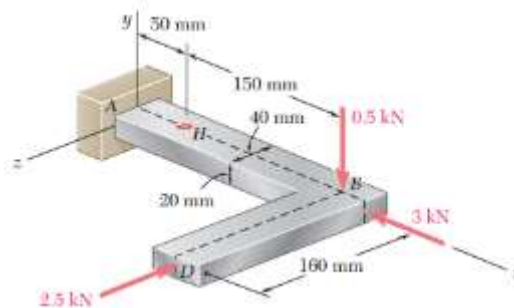


- A ductile hot-rolled steel bar has minimum yield strength in tension and compression of 350 MPa. Using the distortion-energy and maximum-shear-stress theories determine the factor of safety for the following plane stress states:
 - $\sigma_x = -60 \text{ MPa}$, $\sigma_y = -85 \text{ MPa}$, $\tau_{xy} = -60 \text{ MPa}$ (given values are plane stress)
 - $\sigma_x = 120 \text{ MPa}$, $\sigma_y = 30 \text{ MPa}$, $\tau_{xy} = -20 \text{ MPa}$ (given values are plane stress)
 - $\sigma_A = 130 \text{ MPa}$, $\sigma_B = -40 \text{ MPa}$ (given values are principal stress)
- Repeat Prob. 4 by first plotting the failure loci in the σ_A, σ_B plane to scale; then, for each stress state, plot the load line and by graphical measurement estimate the factors of safety.

6. Two forces are applied to the pipe AB as shown. Knowing that the pipe has inner and outer diameters equal to 35 and 42 mm, respectively, determine the normal and shearing stresses at (a) point a , (b) point b .

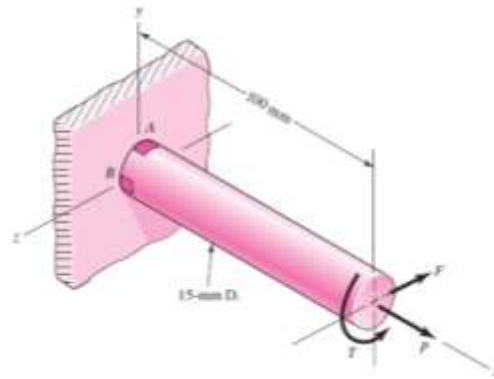


7. Three forces are applied to the machine component ABD as shown. Knowing that the cross section containing point H is a 20X40-mm rectangle, determine the principal stresses and the maximum shearing stress at point H (use Mohr's circle).

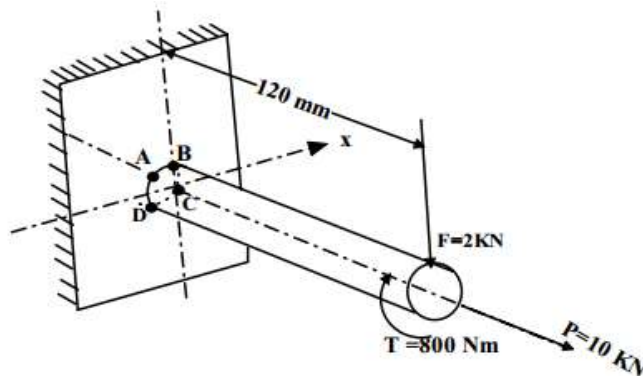


8. A 2" diameter steel bar with $S_y = 50$ ksi is under pure torsion of a 20,000 in-lb. Find the factor of safety guarding against yielding based on:
- Distortion energy theory, and
 - Max shear stress theory.
9. A circular shaft, 100 mm diameter is subjected to combined bending moment and torque, the bending moment being 3 times the torque. If the direct tension yield point of the material is 300 MN/m^2 and the factor of safety on yield is to be 4, calculate the allowable twisting moment by the following theories of failure:
- Maximum principal stress theory
 - Maximum shear stress theory
 - Distortion energy theory
10. A horizontal shaft of 75 mm diameter projects from a bearing and, in addition to the torque transmitted, the shaft carries a vertical load of 8 kN at a 300 mm from the bearing. If the safe stress for the material, as determined in a simple tension test, is 135 MN/m^2 find the safe torque to which the shaft may be subjected using as a criterion
- The maximum shearing stress
 - The distortion energy

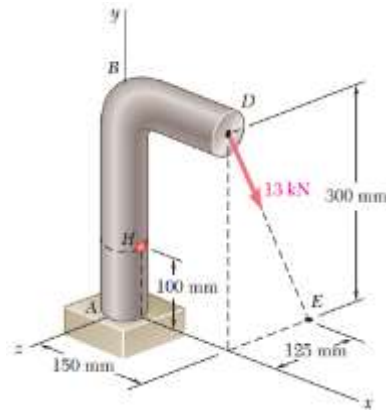
11. A structure is composed of circular members of diameter d . At a certain position along one member the loading is found to consist of a shear force of 10 kN together with an axial tensile load of 20 kN. If the elastic limit in tension of the material of the members is 270 MN/m^2 and there is a factor of safety of 4, estimate the magnitude of d required according to (a) the maximum principal stress theory, and (b) the maximum shear strain energy per unit volume theory. Poisson's ratio $\nu=0.283$.
12. This problem illustrates that the factor of safety for a machine element depends on the particular point selected for analysis. Here you are to compute factors of safety, based upon the distortion energy theory, for stress elements at A and B of the member shown in the figure. This bar is made of AISI 1006 cold-drawn steel and is loaded by the forces $F = 0.55 \text{ kN}$, $P = 4.0 \text{ kN}$, and $T = 25 \text{ N.m}$.



13. A cantilever rod is loaded as shown in the figure shown. If the tensile yield strength of the material is 300 MPa determine the rod diameter using
- Maximum principal stress theory
 - Maximum shear stress theory
 - Maximum distortion energy theory



14. A 13-kN force is applied as shown to the 60-mm-diameter cast-iron post ABD. At point H, determine (a) the principal stresses and principal planes (use Mohr's circle), (b) the maximum shearing stress.



15. A cast aluminum 195-T6 exhibits $S_{ut} = 36$ kpsi, $S_{uc} = 35$ kpsi, and $\epsilon_f = 0.045$. For the given state of plane stress, (a) using the Coulomb-Mohr theory and modified Mohr theory, determine the factor of safety, (b) plot the failure locus and the load line, and estimate the factor of safety by graphical measurement.
16. A round cantilever bar made of brittle material experience torsion applied to the free end. Assume that the compressive strength is twice the tensile strength. Express failure stress in terms of strength.

Use:

- The polar moment of inertia, $J = \frac{\pi c^4}{2}$
- The moment of inertia, $I = \frac{\pi c^4}{4}$ where c is radius of the circle